

IB Mathematics - Area Between Two Curves

10 Paper 2 Style Exam Questions | AAHL & AIHL

Instructions: Give exact answers where appropriate. Otherwise give answers to the required degree of accuracy. Show all relevant working. Calculator use is expected where numerical methods are requested.

Question 1 [7 marks]

The curves $y = x^2 - 2x$ and $y = 2x - x^2$ enclose a finite region.

- (a) Find the coordinates of the points of intersection. [2]
- (b) Hence, find the exact area enclosed by the two curves. [5]

Question 2 [8 marks]

Consider the curves $y = x^3 - 3x$ and $y = x$.

- (a) Find the x-coordinates of all points of intersection. [3]
- (b) Sketch the two curves on the same set of axes, indicating the points of intersection. [2]
- (c) Find the total area of the regions enclosed by the curves. [3]

Question 3 [8 marks]

The graph of $y = \ln(x)$ and the line $y = x - 1$ intersect at $x = 1$.

- (a) Show that $x = 1$ is the only point of intersection for $x > 0$. [2]
- (b) Find the area of the region bounded by $y = \ln(x)$, $y = x - 1$ and the y-axis. [6]

Question 4 [9 marks]

The curves $y = e^x$ and $y = 2x + 1$ form a bounded region.

- (a) Use a graphical or numerical method to find the two x-coordinates of intersection, giving answers correct to 3 significant figures. [3]
- (b) Determine which function is greater between the two intersection points. [1]
- (c) Hence, find the area of the bounded region, giving your answer correct to 3 significant figures. [5]

Question 5 [8 marks]

Let $f(x) = 4 - x^2$ and $g(x) = x + 2$.

- (a) Find the coordinates of the points where the graphs of f and g intersect. [3]
- (b) Find the area enclosed between the graphs of f and g . [5]

Question 6 [9 marks]

The curves $y = \sin(x)$ and $y = x/2$ enclose a region for $0 \leq x \leq \pi$.

- (a) Find the x-coordinate of the second point of intersection in the interval $0 < x < \pi$. Give your answer to 3 significant figures. [3]
- (b) Set up, but do not evaluate, an integral representing the area between the curves from $x = 0$ to this point. [2]
- (c) Use your calculator to find the area, giving your answer to 3 significant figures. [4]

Question 7 [10 marks]

The curve $y = \sqrt{x}$ and the line $y = x/2$ enclose a finite region.

- (a) Find the x-coordinates of the points of intersection. [3]
- (b) Explain why the area should be integrated with respect to x rather than y for the most direct method. [1]
- (c) Find the exact area of the enclosed region. [6]

Question 8 [9 marks]

The curves $y = x^2$ and $y = 2\sqrt{x}$ intersect at two points in the first quadrant.

- (a) Find the coordinates of the intersection points. [3]
- (b) Find the exact area enclosed by the curves. [6]

Question 9 [10 marks]

A region is bounded by the curve $y = 6 - x^2$, the line $y = 2x + 2$, and the x -axis.

- (a) Find the x -coordinate of the point where the curve and line meet. [2]
- (b) Find the x -intercepts of the curve. [2]
- (c) By considering the relevant intervals, find the total area of the region. [6]

Question 10 [11 marks]

The functions $f(x) = x^3 - 4x$ and $g(x) = 2x$ are shown on the same axes.

- (a) Find the x-coordinates of the points of intersection. [3]
- (b) Explain why there are two separate enclosed regions. [2]
- (c) Find the area of each enclosed region. [4]
- (d) Hence, find the total area enclosed by the two curves. [2]
